Bibliographic support of the extended formula, by component and methodological decision, and type of support

Table 9 of the manuscript. Reference numbers follow the IEEE numbering of the manuscript (order of first citation). Type of support: C, conceptual precedent for the element; M, methodological support for a specific modeling choice; D, data or candidate input; A, modeling choice of the authors.

Component or decision	Sources	What they support in the model	Type
Power and its instruments	[3]–[7]	Definition of power and of cyber power; the four instruments of national power; naval power as the expression of military power at sea; the diplomatic, information, military, and economic influence of maritime forces; maritime power as the capacity to create, develop, exploit, and defend maritime interests	C
Foundations of maritime power	[4], [6], [7], [10], [18], [20], [21]	Elements of sea power; maritime strategy beyond the fleet; the military, diplomatic, and policing functions of navies and their loss-of-power gradient, which cyberspace escapes in reach but not in effect, since location still matters as a resource in the cyber domain; maritime power as a system of interconnected elements	C
Classical core MP = MI × NP	[8], [9], [19], [22]–[24]	Maritime power as the product of maritime interests and naval power, its Colombian adoption, and the content and analytical limits of the formula; the institutional description of maritime power as a sum	C
Colombian institutional framework	[1]–[3], [23]	Maritime interests as a factor of national development and security; the national ocean policy as the frame of the national and sectoral development plans, with position, political will, and maritime interests as the variables of its goal; the protection of maritime interests through naval power as a strategic objective	C
MI	[3], [9], [10], [23]	Purpose of maritime power; n = 18 maritime interests for Colombia, which, unlike the thirteen of the 2018 policy, exclude naval power; identification of interests as a prior condition of naval planning	C, D
$NP = (F \times P \times MDA^{\mu})^{1/(2+\mu)}$	[9], [10], [16], [18]	Force and position; technology and doctrine; Colombian naval power; MDA elevated to the level of F × P	C
Cyberdefense in F and in CD_m; boundary with CS_m	[30], [52]	Operational and support-and-sustainability groups of the MARCIM capability taxonomy; the two lines of defense	C
SW	[3], [9], [53], [54], [57]	Will of the state as a constituent of maritime power; political will as a variable of the national ocean policy; political will as the lever for exploiting maritime interests; maritime consciousness in the maritime power system	C
PW, IW, SocW	[55], [56], [58]	Institutional effectiveness, cohesion, and popular support in the national-will literature; inclusion of will with a declared range of uncertainty	C
Will as a multiplier of capability	[44], [45]	Capability multiplied by intent; political will as a principal and rarely observable determinant of cyber power	C, M
MS	[17], [25], [59]	Three threat dimensions (IC, TE, BC); the security regime of ships and port facilities; the maritime security rank of Colombia	C, D
CP as a structural variable	[4], [12], [16], [27], [38], [39], [43], [46], [47]	Definition of cyber power; hard and soft cyber power; cyber power as a key element of maritime power; cyber dimensions of sea systems and the balance of cyber dependence; cyber dependence of the Colombian maritime domain; vulnerabilities of critical maritime systems; cyber power as an articulation of technical, normative, defensive, and diplomatic capabilities; conceptual links between cyberspace and naval power	C
MARCIM framework	[29], [30], [40]	Reference framework for the modeling and simulation of maritime cyberdefense, its simulation model SERDUX-MARCIM, and the doctoral dissertation in which both were developed	C
CS_m	[26], [37]–[39]	Breaches associated with operational technology; interconnection of shipboard IT and OT systems; vulnerabilities of AIS and ECDIS; cyber risk management in safety management systems	C
RC_m	[4], [30], [60]	Recovery rate of SERDUX-MARCIM; deterrence by futility, also the rationale for the additive aggregation of CP; national diagnosis of crisis management and incident response	C, D
CDip	[28], [61]	Cyber diplomacy as an instrument of projection; the United Nations framework of responsible state behavior and the Open-Ended Working Group	C
Cyber context and incidents	[11], [13]–[15], [41], [42]	Digital dependence of ports; the 2017 Maersk attack and the 2019 incident reported by the U.S. Coast Guard; cyberspace as a strategic domain; data-driven cyberdefense	C
Normalization and aggregation	[48]–[51]	Weighted geometric mean; limited substitutability as the criterion for choosing between linear and geometric aggregation; weights as trade-offs; dependence of a component's influence on its variance and correlations	M
Verification, validation, and robustness	[48], [62]–[64]	Definitions of verification, calibration, and validation; extreme-condition, degenerate, and sensitivity tests; Monte Carlo uncertainty analysis with Sobol' indices and expert-weight sampling	M
Existing measures of maritime and naval power	[17], [31]–[36]	Composite measures in Latin America and beyond: all additive, none including cyber power; the limits of shares of a system total; candidate benchmarks for external validation	C, M, D
Measures of cyber power and candidate inputs	[44]–[46], [60], [65], [66]	National cyber power measures without a maritime dimension; indices proposed as inputs for CS_m and CD_m	C, D
Modeling choices of the authors	—	Functional forms of the sub-indices (Equations 4–8) and the normalization of naval power; the reference ranges of γ, α, μ, and δ; the lower bound of the rank transformation; the five-level thresholds and the five bands of CP. They are not taken from a source: the rationale for the aggregation rules is given in Section IV-D and that for the ranges of γ and α in Section IV-A; Proposition 1 proves the properties of the Level 1 formula; Section V and Tables 7 and 8 test the ranges, the Level 2 weights, the aggregation rules, the rank transformation, and the goalposts of the indicators; the level thresholds and the bands of CP remain conventions until calibrated	A